

Palak Singh

✉ palak.singhhh23@gmail.com ☎ +91 7004993499 in Palak 🌐 Palak </> LeetCode

Education

Vellore Institute of Technology

B.Tech in Computer Science - 8.83GPA

Bhopal, MP

Sept 2023 – May 2027

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management, Computer Networks

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, SQL, C++

Computer Science: Data Structures, Algorithms, System Design, Object-Oriented Programming, Distributed Systems

Backend & APIs: Node.js, Spring Boot, FastAPI, REST, GraphQL, Microservices, Kafka

Frontend: React.js, Next.js, TypeScript, Tailwind CSS, Redux, Zustand

Databases & Caching: PostgreSQL, MongoDB, Redis, Query Optimization

DevOps & Cloud: AWS (EC2, S3, EKS), Docker, Kubernetes, Terraform, CI/CD, Git

Experience

Software Development Intern

MyTripGoals

Noida, UP

Aug 25 – Current

- Engineered scalable **travel booking platform** using **React, Node.js, and PostgreSQL**, streamlining booking operations and improving agent productivity by **40%**.
- Developed modular frontend architecture with **TypeScript, Tailwind CSS, and reusable component systems**, accelerating feature delivery and maintaining design consistency across dashboards.
- Implemented **real-time communication infrastructure** using **Socket.IO, Redis pub/sub, and horizontal scaling strategies**, reducing cross-instance message latency by **15%**.
- Integrated **RESTful APIs**, caching layers, and backend workflows for scalable booking and payment systems.
- Contributed to **CI/CD workflows**, code reviews, and automated testing with **Jest** to improve deployment stability.

Projects

EduCore - School ERP SaaS Platform

[Link](#)

- Architected **multi-tenant SaaS ERP** using **Node.js & TypeScript**, powering 40+ modules across multiple schools.
- Served **4K+ active users**, optimized with **Redis caching**, database indexing, and efficient backend workflows, ensuring fast, scalable, and highly reliable system performance during peak usage.
- Integrated **Razorpay** with **PostgreSQL**, ensuring ACID-compliant transactions and automated reconciliation.
- Processed **6L+ transactions** across **2,500+ payments** with **99.9% success**, implementing idempotent payment handling, retry mechanisms, and transaction validation to maintain reliability and consistency.

SafeSurf - Real-Time Mobile Content Moderation System (Government-Funded)

[Link](#)

- Implemented real-time **content filtering pipelines** with low-latency inference for instant unsafe media detection.
- Developed Android client using **Java and Kotlin** in Android Studio, integrating real-time content moderation APIs.
- Built **Spring Boot backend** leveraging **TensorFlow and NLP models** to detect NSFW content across mobile applications.
- Made comprehensive **JUnit and Mockito test suites**, achieving 85%+ coverage and reducing production defects.

Achievements

Competitive Programming: Solved 850+ problems on LeetCode (Rating: 1700+, Top 8%) — 200-day active streak

Hackathon Recognition: 5× Winner at CodeVerse, HackByte, Cisco Codeathon, Hack4Health, MP Police Cyber Security

Google Summer of Code 2024: Selected as contributor (Top 2% globally), Contributed to open-source projects

Technical Leadership: Trained 200+ students through 15 AWS/Azure workshops as community leader

Cloud Certifications: Azure Fundamentals (AZ-900)

Freelance Experience: Delivered 6 production applications ([SDPL](#), [Plotify](#), [EduCore](#)) serving 20,000+ users

Publications & Authorship

Research Publications: Published 3 papers including "[Leveraging Kubernetes for LLM Deployment](#)" in IJFMR (Jan 2025)